

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 8.4 returns to the

23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observ-
44 able onset. Coherent adaptation can lose alignment with a designated en-
45 vironmental template, and a compatibility estimator needs diagnostics that

distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Sections 2–4 develop the three requirements together (Fig. 1). Section 5 composes them, stating the thermodynamic constraints and all eight connecting hypotheses per branch; its supporting models of feedback, learning and resource supply have distinct state spaces and assumptions. Section 6 then examines what the observables can establish, before Section 7 develops the cortical hypothesis and an awakening experiment. The hypothesis is that extracellular contraction increases field-mediated coupling; its direction, rate and relation to recovery require independent calibration. Supplemental Material organizes the constructions and controls behind the claims they support.

2 Coherence: physical coupling and its resource condition

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

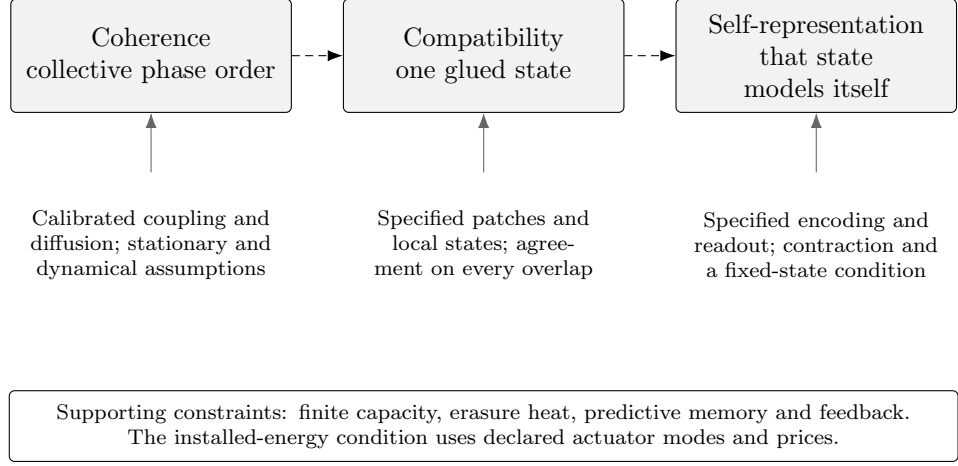


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

70 where I_n are modified Bessel functions and ψ is the collective phase. The for-
71 mal derivation uses the absence of stationary current for a periodic gradient
72 drift to obtain this density, then verifies normalization and self-consistency.
73 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one
74 positive solution coexists with zero and increases strictly with K . Its onset
75 is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

76 Figure 2 shows this stationary branch.

77 Linearization about the uniform density gives rate $K/2 - D$ for the first
78 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
79 exactly above threshold. About the coherent density, the quadratic form
80 is strictly dissipative transverse to collective rotation, whose rate is zero.
81 These linear estimates concern the limiting equation. Relating them to finite
82 stochastic trajectories requires a dynamical mean-field limit and branch selec-
83 tion; exchange symmetry alone does not establish the needed independence
84 (Sznitman, 1991).

85 **2.2 Installed energy bounds mean coupling**

86 A spatial model replaces the finite sum by integration against a kernel. One
87 declared implementation carries response modes with continuous profiles ϕ_i ,

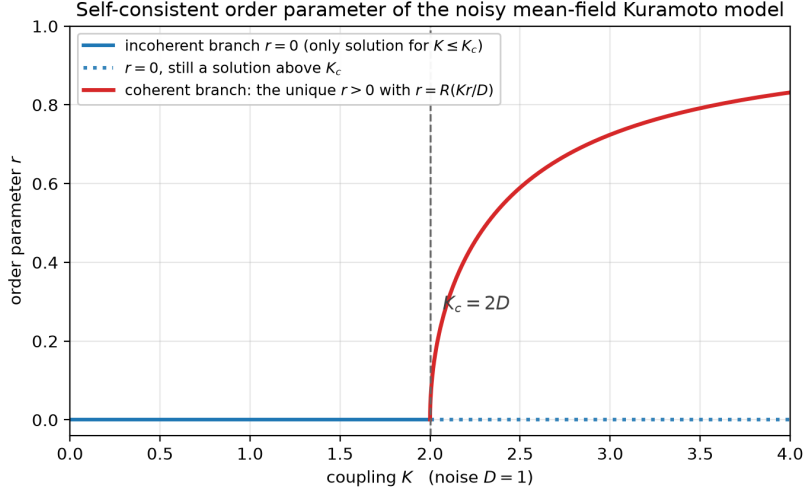


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

88 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

89 The same configuration sets the reciprocal kernel and stored energy. An
 90 executed transition changes the kernel through its changed occupancies and
 91 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 92 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 93 pair reconstruction on refining spatial partitions converges to its product-
 94 measure integral under the specified compactness and sampling assumptions;
 95 the refinement index is independent of the number of control updates.

96 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 97 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 98 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \tag{5}$$

99 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 100 coherent stationary branch and any growing Fourier mode about the uniform
 101 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 102 is necessary: energy above it need not produce sufficient coupling because
 103 the bound can be loose.

104 The quantity constrained is installed energy. Dissipated heat does not
 105 determine it, and κ is supplied by the mode profiles, prices and substrate
 106 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 107 tial dynamics requires a justified reduction. Identifying the installed kernel
 108 with a cortical field is a separate physical commitment.

109 **3 Compatibility: from coordinated activity** 110 **to a global description**

111 **3.1 Why common timing is insufficient**

112 For overlapping patches U and V , compatibility means that restricting either
113 local description to $U \cap V$ gives the same object. Descriptions can agree on
114 shared quantities while contributing different information elsewhere. Com-
115 patible sections of the specified sheaf have a unique global extension. On
116 a finite cover, local finite spatial measures agreeing as measures on every
117 overlap also have a unique finite-measure extension whose sheaf image is the
118 glued section. No mass-one normalization is imposed.

119 In the cup example, visual and reaching descriptions must express a
120 shared quantity, such as cup location, in common coordinates. Restriction
121 retains that quantity while dropping details particular to each description.
122 Agreement permits surface appearance and hand configuration to contribute
123 to one situation. The mathematical counterexample assigns the same phase
124 to every patch while two local mass profiles disagree on a shared site. Those
125 profiles have no common extension. Because they are chosen independently
126 of phase, synchrony places no constraint on their disagreement.

127 This distinction limits the biological interpretation. Receptive territories
128 and content domains must be related to spatial patches, and the descriptions
129 must support the required extension. Matching overlap marginals of de-

130 coded probability laws need not determine a unique global joint distribution
 131 (Abramsky & Brandenburger, 2011).

132 **3.2 When coherence does constrain content**

133 A declared relation between phase and content gives a positive result. Let
 134 contents take values in a real normed vector space E . Patch i reads an
 135 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 136 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 137 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 138 sites; their equality on overlaps is a supplied relation between descriptions of
 139 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 140 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 141 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

142 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 143 agree exactly. The factor N makes this estimate potentially weak in large
 144 populations; it supplies a pointwise guarantee from a population average. A
 145 shared scalar encoder is a special case. Dropping the overlap relation permits
 146 disagreement even at perfect locking. A common nonexpansive linear projec-
 147 tion preserves agreement, but agreement of a projection does not establish
 148 agreement of the full vector.

149 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 150 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 151 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 152 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

153 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 154 duce an initial mismatch. Approximate observations leave a nondecaying
 155 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 156 update to the phase process.

157 **3.3 What approximate agreement determines**

158 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 159 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 160 Weights that vanish outside their patches and sum to one at each site select
 161 a global profile by weighted averaging. The selection lies within ε of each
 162 patch there, and any two such selections differ by at most ε , independently
 163 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 164 $L\varepsilon$ across selections.

165 At zero tolerance, the selection is the unique exact extension. At every
 166 positive tolerance there are families with no exact extension; selection then
 167 gives an approximation whose dependence on weights is bounded. Supple-
 168 mental Material gives a three-site witness attaining the bound. The mass-

169 profile construction uses the scalar case of the content dynamics and addi-
 170 tionally requires nonnegativity of the initial profiles and observations. The
 171 general vector result supplies no identification with the spatial-measure sheaf.
 172 These constructions concern declared objects: a specified cover, contents
 173 valued in a normed space, and encoders related on overlaps. They select
 174 neither the neural cover nor the decoder. Identifying their contents with
 175 perceptual content, and establishing that the identification bears on experi-
 176 ence, is a further modelling task; Section 6 sets out what a measurement of
 177 it would have to establish.

178 **4 Self-representation: a state that models it-** 179 **self**

180 **4.1 The reflexive construction**

181 Compatibility gives a global state. Reflexivity relates it to an encoding within
 182 that state. Equip global sections with a complete metric. If the additional
 183 restriction-resonance condition identifies encoding with restriction to a region
 184 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \tag{8}$$

185 The region and readout are fixed across the state space. If F contracts and
186 fixes the section s_* formed from the local descriptions, Banach's theorem
187 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
188 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
189 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
190 one for $\tau > 0$. Constructing a map with that factor remains a modelling
191 obligation.

192 In the cup example, the global state includes the represented scene and the
193 perceiver's relation to it. A local encoding must retain enough information
194 for the readout to recover that modelled state; cup position alone would be
195 insufficient if hand configuration and viewpoint could not be reconstructed.

196 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
197 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

198 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
199 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
200 weakens.

201 4.2 Reconstruction across a family of states

202 Sensitivity can be required quantitatively. Declare relevant macrostates, an
203 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that

204 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

205 Thus M relevant states pairwise more than 2ε apart require at least M
 206 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 207 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 208 existing self-prediction map.

209 This is an additional accuracy criterion on a family, with no requirement
 210 for multiple exact fixed points. A blind encoder or constant readout fails it
 211 for separated states despite having a unique fixed point.

212 4.3 Agreement after a change takes communication

213 Snapshot compatibility does not specify how quickly descriptions respond
 214 when a shared quantity changes. In a synchronous network whose updates
 215 read only the site's own state and declared neighbours' messages, two initial
 216 configurations agreeing on a site's causal past through T rounds produce the
 217 same state there at round T . If two fresh interventions outside that past
 218 require reports separated by more than 2ε , no report from that site can be
 219 within ε of both. This follows from the reconstruction bound in Section 4.2.

220 A three-site delay line witnesses failure after one round and exact report-
 221 ing after two; cutting the path prevents delivery at every round. A report
 222 may still happen to be correct about one fixed world. The bound concerns

223 responsiveness across interventions.

224 Throughout this section *prediction* names a reconstruction map; reading
225 it as temporal neural prediction would require dynamics. The relevant family
226 of states, the metric, the tolerance, the communication graph and the dead-
227 line are all declared inputs, as are the represented variables and the fidelity
228 counted as accurate. Reading the resulting code count as physical memory
229 capacity requires a hardware model, and a readout that varies with the state
230 requires a further one.

231 **5 Thermodynamics and composition**

232 The phase and representation results concern coupling, content disagreement
233 and reconstruction error. Thermodynamics constrains the physical opera-
234 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
235 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
236 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
237 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
238 composition connects these objects through the explicit hypotheses in Ta-
239 ble 1.

240 5.1 Feedback changes the information constraint

241 For passive memory X_t driven by an environmental signal S_t , the predictive
242 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

243 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
244 et al., 2012). Zero retained information also satisfies it, so a task and a
245 learning rule require additional specification. Relating this bound to Lan-
246 dauer heat requires a common thermal scale and budget, and memory waste
247 bounded by erased entropy.

248 An acting agent changes the signal it will receive. Specify policy, world,
249 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \quad (12)$$

250 The second conditional term vanishes in the passive model. With feedback,
251 the signed difference can be negative as action creates correlations with the
252 future. A causal reading requires an environmental state sufficient to describe
253 the relevant dynamics.

254 For a controlled substep, the active thermodynamic bound instead uses
255 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
256 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
257 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,

258 2013). Every forward path of positive probability must have a positive-
259 probability reverse for this finite identity. A missing reverse gives infinite
260 divergence. The supplement develops the support conditions and composes
261 actuation with memory updates on their shared intermediate law.

262 **5.2 Learning and repeated operation need explicit re-** 263 **sources**

264 The finite examples distinguish selecting a feasible policy, relaxing a register
265 toward an encoded reward, and learning an unknown task from observa-
266 tions. The observational learner acquires correlation with an environmental
267 parameter while its action marginal stays uniform. A continuing model exe-
268 cutes actuation, observation-driven learning and reset in sequence, preserving
269 above-chance performance at a positive work cost per cycle.

270 A phase-derived sensor makes the informational distinction concrete: an
271 informative readout and a blind control have the same phase order but dif-
272 ferent learning performance. Attaching the learner’s register to the actuator
273 of Section 2.2 ties reward, installed kernel and work to its own executed law.
274 The learning witness rises above the necessary installed-energy bound while
275 its matched control remains on the exclusion line. These finite constructions,
276 detailed in Supplemental Material, leave the evolving feedback between phase
277 dynamics, installation and sensing to be specified.

278 For that model, common energy E , thermal scale θ , local detailed balance

279 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

280 where expected cumulative work stays within the supplied allowance B at
 281 every prefix. This bounds expectations. Separate constructions track a store
 282 on each trajectory, include finite sources, charge sensor clearing and prepend
 283 preparation. They specify when a mean allowance represents an available
 284 resource; an expected budget alone permits overdrawn paths.

285 These constructions establish costs for their declared processes. A finite
 286 source with a positive net cost permits only finitely many funded cycles,
 287 while continuing operation requires replenishment. Register–bath witnesses
 288 also distinguish actual bath-energy transfer from transition-log heat. Ident-
 289 ifying a work store, an erasure budget and the installation energy of Sec-
 290 tion 2.2 requires physical relations that these separate models do not supply
 291 automatically.

292 **5.3 What the composition establishes**

293 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 294 cept eight connecting implications (Table 1). In the active branch, a named
 295 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

296 with B the register’s heat. The upper allocation follows from a ledger whose
 297 operations deliver that total heat to one reservoir and whose other operations
 298 do not increase joint entropy. Those ledger assumptions are physical inputs.
 299 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Joint witnesses combine a register, boundary, refining mesh and small spatial cover; active witnesses add controlled thermal updates and a priced installation. They establish satisfiability under their stated identifications. Mesh convergence follows from refinement data, and the witnesses establish no common cortical mechanism. Supplemental Material maps the claims to their Lean declarations and records the counterexamples. The development

312 declares no additional axioms beyond its libraries' foundational axioms.

313 6 What the observations establish

314 The composition holds under its premises. Whether observation can estab-
 315 lish those premises is a separate question, and the three requirements it
 316 connects concern different objects: phase dynamics, local descriptions and
 317 internal encodings. Their observables must retain these distinctions. The
 318 following controls show which mechanisms remain indistinguishable, which
 319 temporal approximations fail, and what independent content measurements
 320 must establish.

321 6.1 Stationary distributions leave the mechanism un- 322 determined

323 Equation (2) gives a concentration-coherence relation with no fitted shape
 324 parameter. Estimate r from circular resultant length and a from density
 325 shape: inverting R from the measured r would make agreement automatic.
 326 Even independently estimated agreement identifies a statistical family shared
 327 by different mechanisms.

328 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 329 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 330 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 331 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

332 and diffusion D_0 have that same density. Their stationary distributions co-
 333 incide window by window despite different causes.

334 Resolved dynamics supply additional information. In the ideal model with
 335 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 336 increment means constrain drift. Separating endogenous coupling from a
 337 matched common drive additionally requires a measured drive or controlled
 338 perturbation; identifying electromagnetic coupling requires evidence specific
 339 to that interaction. Supplemental Material gives the equivalence construction
 340 and its numerical check.

341 **6.2 Finite ramps change the observable onset**

342 The square-root law in Eq. (3) describes stationary solutions. A changing
 343 coupling requires time for the phase distribution to approach those solutions.
 344 In a deterministic mean-field control that crosses the threshold symmetrically,
 345 none of the scanned rates meets the declared tracking tolerance: slowing
 346 the crossing does not remove the terminal residual under that initialization.
 347 A slow control variable therefore supplies no general guarantee of branch
 348 tracking at onset. Supplemental Material gives the residual, rate range and
 349 admissible rates away from threshold.

350 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 351 at four coupling speeds. Escape is the first of three consecutive saved samples
 352 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
 353 replicas escape within the fastest window. The three slower ramps have

mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents differ from those of the exact stationary branch evaluated on the same grids. At the fastest speed the ensemble mean never reaches the fit window, so its fitted slope does not estimate stationary onset.

Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002) makes the apparent exponent depend on ramp speed and fitting window. The specified deviation-based critical ramp speed is not bracketed by this sweep. With fluctuating coupling, the relevant stationary comparison also depends on rate: the mean coupling approximates order in the tested fast regime, while averaging stationary order over coupling works for the tested slow drives confined to one side of threshold. Slow crossing drives can satisfy neither approximation. These are finite-horizon observations, with controls and parameters in Supplemental Material.

Additional controls address integration bias, population size and spatial or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ contains 2. These are finite numerical regimes, with full parameters and controls in Supplemental Material; they supply no cortical calibration.

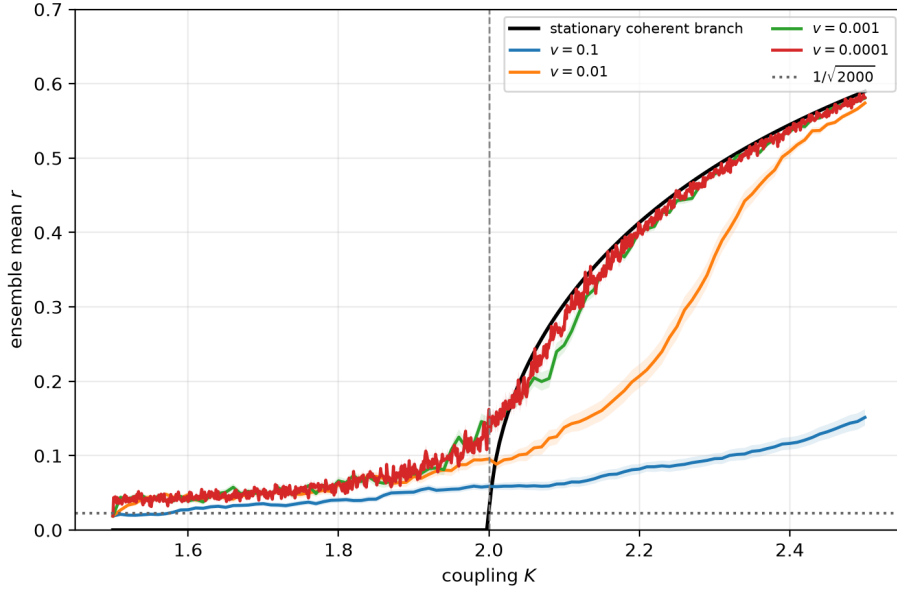


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

372 6.3 A winding state leaves the order parameter am- 373 biguous

374 The order parameter r is indifferent to where a phase sits in space, so a
375 field winding once across a periodic sheet has $r = 0$ while every patch of it
376 stays locked. Such a field is not an artificial construction here: on a torus a
377 twist is stationary for any isotropic symmetric kernel at identical frequencies,
378 because the coupling term is odd in the separation against a kernel even in
379 it. It satisfies the positivity and identical-frequency hypotheses, and violates

only the quarter-turn initial spread that the convergence result additionally requires before placing the limit at the uniform state.

On the sheet of Section 2’s spatial control, a twist survives below an interpolated decay length of 0.1871 mm and unwinds above it, a boundary interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and $r = 0.0083$ from a twisted one, both stationary, with patch-local order 0.9015 and no winding defect. At fixed K the observable therefore takes two values spanning nearly its whole range: K does not determine r , and a low resultant does not by itself distinguish weak coupling from a travelling wave.

Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial control carries, the boundary moves to 0.1421 mm and the same comparison at 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realisations. Heterogeneity shortens the range that sustains a wave without removing it.

This qualifies what a phase trace can establish. An onset criterion stated in r alone registers a change of basin as readily as a change of coupling, so the protocol of Section 7.3 needs a spatial observable beside the resultant. Supplemental Material gives the sweep, its uniform control and the limits of both.

399 **6.4 Coherent adaptation can lose environmental align-** 400 **ment**

401 An adaptive coupling rule might supply representational structure, but its
402 objective must reward the relevant agreement. For symmetric coupling, pair-
403 wise sine terms cancel in the population mean drift. Minimizing squared drift
404 at that fixed mean therefore rewards equalizing drift across nodes. In the
405 frequency-cluster control, this favors coupling between unlike-frequency clus-
406 ters, whereas the diagnostic environmental template rewards coupling within
407 clusters.

408 The formalization proves descent of the squared-drift objective for a spec-
409 ified gradient flow at fixed phases. Its functional differs from thermodynamic
410 entropy production: at a coherent stationary density the probability current
411 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
412 rial gives the operator, its gradient and the conditions of the descent result.

413 The finite control co-evolves noisy phases and symmetric, nonnegative
414 couplings at fixed total resource. Its template is a diagnostic covariance of
415 latent signals, never a training input. Across three seeds, second-half order
416 remains between 0.9167 and 0.9222, while the within-cluster to between-
417 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
418 Random-update, relabelled-gradient and frozen-kernel controls are reported
419 in Supplemental Material.

420 The immediate post-update objective reduction disappears when aver-

421 aged over complete update intervals. A prespecified grid of 6 rate–cadence
422 combinations contains no case meeting the 5% sustained-reduction criterion
423 while retaining coherence and losing alignment. The finding is coherent adap-
424 tation away from the specified structure, with no demonstrated sustained cost
425 reduction.

426 Kernel alignment and overlap compatibility are different observables.
427 This control addresses acquisition of its designated template under its par-
428 ticular objective. Learning models can explicitly include data in an objective
429 (Corberi et al., 2026); an account of agreement about the cup must specify
430 the content its learning rule is meant to preserve.

431 **6.5 Content agreement needs independent decoders**

432 A neural test needs independently identified domains, local descriptions and
433 restriction maps. Two decoders should estimate a shared content on overlap-
434 ping domains before their agreement is examined. Discrepancy requires a null
435 under decoding noise and controls using deliberately compatible and incom-
436 patible examples. An implementation on constructed data separates these
437 cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture
438 agreement, and regional bias can manufacture disagreement. Per-patch de-
439 coding diagnostics detect these failures; the compatibility statistic alone does
440 not.

441 Phase agreement cannot replace that measurement. On constructed con-
442 tent, phase carries a share 1.000 of the content entropy when content is built

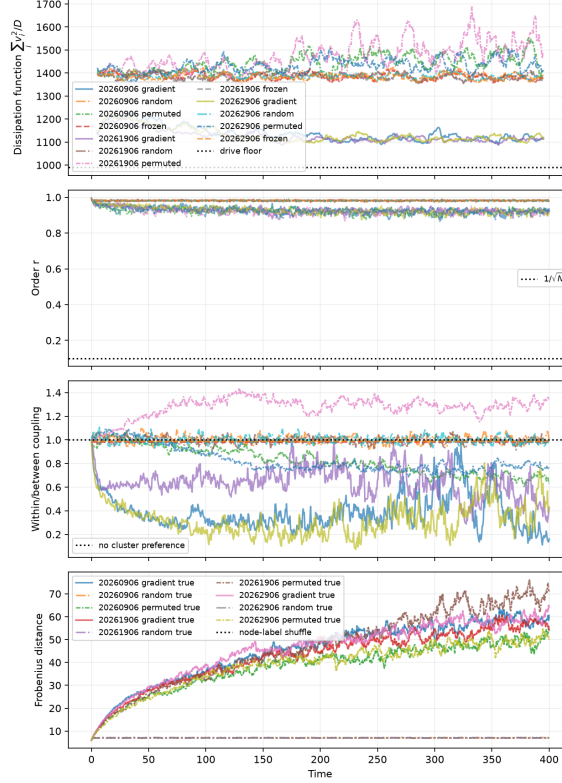


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

443 from it and 0.000 when it is not. A phase-derived statistic reports com-
444 patibility on phase-locked patches whose local descriptions disagree. The
445 implemented content statistic detects their incompatibility. These controls
446 validate a measurement on known constructions; a neural content model and
447 decoder remain to be established.

448 Choosing the cover also chooses a candidate subject: trivial descriptions
449 can agree while carrying little content, and a partition can conceal disagree-
450 ment. The useful choice must therefore be constrained by what neural ac-
451 tivity encodes. The exact composition assumes compatibility of its supplied
452 cover. The content mechanisms above identify additional hypotheses under
453 which agreement can arise; they leave the empirical choice of content and
454 the connection to that cover explicit.

455 **6.6 Phase-density estimation and measurement feasi-** 456 **bility**

457 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
458 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
459 tages produce different residuals. The bipolar results are compatible with the
460 Bessel relation, but their concentration range cannot distinguish it from its
461 linear approximation at the observed scatter. Temporal pooling changes the
462 estimand, and sedation recordings leave the awakening hypothesis untested.

463 The concentration range and dependence determine whether the relation

464 can be tested. Under the specified log-density estimator and bias-floor crite-
465 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
466 3000 independent sites in the scanned designs, rising to 31000 when sites
467 share a mean in groups of 10. These counts describe that estimator and de-
468 pendence model; the intended recording requires its own calibration. Supple-
469 mental Material reports the bin-count controls and the full dependence-aware
470 design.

471 An informative dataset needs sufficient concentration range or precision,
472 calibrated estimators and an observation model for dependence, source mix-
473 ing and reference choice. Public mouse calcium imaging provides thousands
474 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
475 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
476 here. Site count, state coverage and bandwidth are separate requirements.

477 **7 The cortical hypothesis and an awakening** 478 **test**

479 The cortical proposal identifies a physical source of coupling and asks whether
480 its changes help explain recovery. The tests in Section 6 determine what
481 observations must distinguish before a phase trace can support this identifi-
482 cation.

483 7.1 Physical identification and calibration

484 The cortical hypothesis identifies part of effective coupling with endogenous
485 extracellular electromagnetic interactions. Fields can modulate spike timing
486 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
487 propose coordination of distributed neural information (Pinotsis & Miller,
488 2022; McFadden, 2020). Phase organization under propofol motivates study-
489 ing this regime (Bastos et al., 2021), but does not measure coupling.

490 The formal field predicate requires a jointly continuous kernel, a normal-
491 ized substrate, coupling equal to the kernel average, and no single site pro-
492 viding more than half that strength. Toy constructions satisfy it. A cortical
493 realization must specify the scale at which the kernel and its reduction ap-
494 ply: calcium imaging reports different state-dependent modularity patterns
495 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

496 The spatial reduction is also testable. A row-sum aggregation at zero
497 diagonal tracks the uniform-coupling control over the tested spatial decay
498 range, while frequency heterogeneity displaces the operational threshold. The
499 fit window biases both estimates relative to $2D$; agreement with the control
500 is therefore the supported conclusion. Supplemental Material reports the
501 reduction, its numerical error and the distinct assumptions needed for the
502 kernel-average identification in Eq. (5).

503 The field commitment requires physical evidence beyond a choice of math-
504 ematical representation. A finite matrix spread over positive-mass cells pre-
505 serves its total coupling and phase correlation exactly. Point-supported

506 vanishing under an atomless measure and a disadvantage from excluding
 507 a better-correlated pair concern specified support constraints. Neither com-
 508 parison supplies energy prices or a biological–digital distinction.

509 A spike-time shift alone supplies no coupling rate. The dimensionless
 510 product of population size N , timing sensitivity s , field amplitude E and
 511 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

512 This rate describes how induced timing shifts accumulate into drift. Nei-
 513 ther it nor the installation factor κ follows from a heat bound. Testing the
 514 physical threshold requires independent estimates of the interaction kernel,
 515 its aggregation and diffusion in the same cortical regime.

516 **7.2 Extracellular geometry and recovery**

517 The proposed control variable is extracellular geometry. Extracellular vol-
 518 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 519 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 520 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 521 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 522 an effective-medium description predicts that a larger extracellular fraction
 523 raises conductivity and reduces field amplitude.

524 The coupling-gated hypothesis consequently predicts that extracellular

525 contraction on awakening raises E and, holding the other factors in Eq. (15)
 526 fixed, raises effective coupling. This argument supplies a direction; the
 527 geometry–coupling relation, its magnitude and its role in human recovery
 528 require measurement.

529 For a smooth crossing at nonzero speed, evaluating the stationary branch
 530 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

531 This follows from Eq. (3) at fixed D and concerns the stationary branch
 532 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 533 dently justified tracking approximation. The crossing controls in Section 6.2
 534 leave that approximation unsupported at onset, even though a separation of
 535 geometry and phase time scales can support it away from threshold. The
 536 awakening hypothesis therefore needs a calibrated dynamical prediction for
 537 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 538 does not supply a recovery latency.

539 **7.3 A protocol for the awakening experiment**

540 The physical test has six requirements:

- 541 1. **Coupling and diffusion in the same units.** Estimate them from re-
 542 solved phase dynamics, separating heterogeneity and observation noise.
 543 Measure the drive or perturb the system to distinguish endogenous cou-

- 544 pling.
- 545 2. **Geometry calibrated to coupling.** Validate the geometry observ-
546 able and its relation to effective coupling, with uncertainty. The con-
547 ductivity argument supplies only the expected direction.
- 548 3. **Calibrated phase measurements.** Estimate r by resultant length
549 and a by unweighted regression of log density on centered cosine.
550 Choose site count and binning from a dependence-aware calibration
551 over the intended concentration range. The tested designs require
552 thousands of sites at the observed concentrations; temporal pooling
553 changes the spatial estimand.
- 554 4. **An independent recovery endpoint.** Measure behavioral recovery
555 separately from phase organization.
- 556 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
557 old above the finite-size floor and evaluate its latency under the mea-
558 sured coupling trajectory, including ramp speed or fluctuations. Record
559 a spatial observable beside the resultant, since a winding field reports
560 a low resultant at coupling that sustains local order (Section 6.3).
- 561 6. **Matched model comparison.** Compare coupling increase, diffusion
562 change, common drive and alternative recovery models under the same
563 observation and filtering assumptions.

564 Failure of the concentration–coherence relation over an informative range
565 constrains the density model. Failure of the calibrated geometry–coupling
566 association constrains the physical gate. Better prediction of independent
567 recovery by a competing mechanism counts against its proposed role. A
568 favorable outcome still leaves the content of coordinated activity to be ex-
569 plained.

570 8 Discussion

571 Stated positively, and granting the premises collected in Table 1, the account
572 runs as follows. Extracellular contraction on awakening raises endogenous
573 field amplitude and with it the effective coupling between cortical oscillators.
574 Once that coupling passes $2D$, a coherent phase branch exists and the popu-
575 lation settles onto it. Encoders shared where cortical territories overlap turn
576 that phase order into agreement about the quantities the territories describe
577 in common, so the local descriptions glue: there is one state of which each
578 is a restriction. Part of that state encodes the state itself, and a readout of
579 that part reconstructs it accurately enough, and sensitively enough across
580 the states that matter, to make it the unique state the system models itself
581 as being. On this account the glued state is what makes distributed activity
582 one situation, and its accurate self-reconstruction is what gives that situation
583 a perspective.

584 Every step of that description is conditional, and its conditions are the

585 subject of what follows.

586 **8.1 Unity and a minimal self as empirical commit-** 587 **ments**

588 We propose the state glued from compatible local descriptions as a corre-
589 late of conscious unity, and its accurate internal reconstruction as a correlate
590 of a minimal self. The motivations are that different contents belong to
591 one experienced situation and that the situation is experienced from a per-
592 spective. The conditional results identify relations a realization would need:
593 sufficient installed resources for the specified phase model, agreement about
594 shared content, and an encoding that supports reconstruction at the required
595 fidelity.

596 Compatibility allows different descriptions to contribute different infor-
597 mation. It also permits uncertainty and conflicting judgments: apparent cup
598 position and judged cup position can be distinct quantities in a consistent
599 description of their discrepancy. Which quantities are shared must be de-
600 termined independently. Choosing them solely to secure agreement would
601 deprive the proposal of empirical force.

602 The minimal-self proposal concerns the perceiver’s position and condition
603 within the situation, such as the cup’s location relative to a body that can
604 reach it. The encoding could be implicit and need carry no autobiographi-
605 cal narrative. Its proposed necessity is philosophical: even an unreflective

606 episode has a perspective. Gluing by itself requires no self-encoding, and re-
607 constructability alone supplies no functional role. Altering the encoded hand
608 or viewpoint while holding the external scene fixed should change estimates
609 of the perceiver’s relation to the cup as the readout predicts. Whether that
610 represented perspective is experienced remains a further question.

611 **8.2 What a test of the full proposal must establish**

612 Both uniqueness results concern specified objects. They choose neither the
613 cover nor the encoding, and do not select between candidate subjects. In-
614 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
615 here selection requires independent evidence about neural organization. A
616 constant readout can contract and fix a predetermined state while discard-
617 ing its input. The family criterion in Section 4.2 tests the sensitivity that
618 this construction lacks, provided the relevant states and fidelity are declared
619 independently.

620 A test must separately measure coupling relative to diffusion, content
621 compatibility and self-reconstruction, alongside an independent endpoint for
622 experience. Meeting all three without a unified episode would challenge suffi-
623 ciency. A unified episode with incompatible descriptions would challenge the
624 unity identification; one lacking the specified self-encoding would challenge
625 the reflexive requirement. Awakening, dreaming and dissociative states offer
626 possible settings, but absence of report cannot by itself define absence of
627 experience.

628 The next physical requirement is a calibrated geometry–coupling relation
629 and resolved phase dynamics. The installation bound contributes a nec-
630 essary resource condition where its hardware representation applies; cortical
631 mode prices and field-mediated coupling require measurement. For the repre-
632 sentational connection, an informative experiment asks whether a controlled
633 change in coordination alters independently decoded content agreement, with
634 controls for sensory information and decoder quality. The constructed-data
635 estimator establishes feasibility for specified contents. Neural local descrip-
636 tions, shared overlap encoders and the self-encoding have yet to be identified.
637 The population-size dependence of the phase-to-content bound further limits
638 its quantitative use.

639 The separation results explain why these measurements must be inde-
640 pendent. A stationary distribution leaves several recovery mechanisms indis-
641 tinguishable. Equal phase order can accompany incompatible descriptions
642 or different task information. The tested plasticity rule preserves coherence
643 while losing its designated template. Each finding concerns the specified
644 model and observable; their role is to delimit the evidence needed for the
645 proposed connections.

646 **8.3 GPUs and the limits of no-go claims**

647 For a large language model (LLM) running on graphics processing units
648 (GPUs), the framework requires identifying a physical system whose local
649 descriptions form one state and whose internal encoding reconstructs that

650 same state. The coupling-gated route E56–E78 requires a calibrated physi-
651 cal coupling, its noise regime and a connection from coordination to content
652 agreement. An attention matrix or a shared clock supplies none of these iden-
653 tifications on its own. Computing a field model likewise leaves its realization
654 in the computing substrate to be established. These physical obligations also
655 apply to the cortical proposal.

656 Unity requires the candidate’s local descriptions to agree on indepen-
657 dently identified shared quantities (Section 3). Common timing and fluent
658 output do not establish that agreement. Digital representations can sat-
659 isfy compatibility, however, so finite hardware is compatible with the gluing
660 construction. The empirical commitment is to identify the relevant cover
661 and contents with one experienced situation; mathematical uniqueness alone
662 selects no subject.

663 For self, the system must carry an encoding $E(s)$ and an internal readout
664 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
665 the state produced by gluing. A constant readout shows why a unique fixed
666 point alone is insufficient for informative self-representation. The additional
667 family criterion in Section 4.2 supplies a test: if two relevant macrostates
668 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
669 within error ε . An intervention outside the encoding region’s causal past
670 gives a corresponding obstruction before a declared communication deadline.
671 Identical verbal reports alone do not establish identical internal codes. These
672 conditional no-go results exclude a specified reconstruction at a specified

673 fidelity or deadline, and apply to biological and digital candidates alike.

674 The hardware comparisons in Supplemental Material (“Coupling support
675 and continuum observables”) have similarly restricted scope. Fixed wiring
676 can impose a correlation disadvantage when it excludes a better-correlated
677 pair, but wiring both directions of a perfectly correlated pair permits an opti-
678 mal allocation. Point-supported kernels vanish under an atomless continuum
679 measure, whereas finite positive-mass cell representations preserve discrete
680 coupling weight and correlation exactly. Neither comparison makes the rep-
681 resentation conditions unattainable on GPUs. A universal exclusion would
682 require a necessary condition for consciousness and a proof that every rel-
683 evant GPU implementation violates it; the present conditional composition
684 supplies neither.

685 The neuroscientific case against *human-like* consciousness in the text-only
686 LLMs assessed by Aru and colleagues concerns embodied sensory interaction,
687 thalamocortical organization and biological regulation linking perception and
688 action to survival (Aru et al., 2023). Their absence supports skepticism con-
689 ditional on their causal relevance to experience; those biological mechanisms
690 are not requirements of the abstract gluing and reconstruction definitions.
691 Workspace-like representations reported in some LLMs (Gurnee et al., 2026)
692 provide candidate organization to investigate. Within this framework, the
693 relevant evidence would identify compatible contents and an informative in-
694 ternal reconstruction, or demonstrate their failure under controlled interven-
695 tions.

696 **8.4 Explanatory scope**

697 Integrated-information, global-workspace and predictive-processing accounts
698 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
699 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
700 candidate physical interaction and conditional relations between local content
701 and self-encoding. Gluing and reconstruction permit multiple substrates;
702 the electromagnetic commitment enters through physical calibration. The
703 formalization checks implications under their hypotheses, and its witnesses
704 establish realizability on declared models. They do not together supply a
705 cortical agent.

706 The explanatory question is whether the specified relations occur in neural
707 activity and matter for experience. In the cup example, this means establish-
708 ing how coordination helps maintain one represented situation and how the
709 perceiver’s place within it contributes to the episode. The proposal identi-
710 fies candidate correlates and tests of that identification. Why those relations
711 would be accompanied by experience remains unexplained.

712 **Data and code availability**

713 The Lean development, simulation scripts, tests and saved numerical sum-
714 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
715 `cortical_coherence`. Publication simulation values are generated from
716 saved summaries without rerunning production sweeps. The exploratory

717 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
718 processing and selection are described in Supplemental Material. The repos-
719 itory contains the reproducible analysis and does not redistribute the raw
720 recordings.

721 **Funding and competing interests**

722 This research received no external funding. The author declares no compet-
723 ing interests.

724 **Use of AI tools**

725 AI assistance to this work was substantial and is not separable by task. Sev-
726 eral coding assistants and chat interfaces were used in an interleaved way
727 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
728 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
729 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
730 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
731 accessed with the Hermes CLI. Between them they contributed to the Lean
732 development, the simulation and analysis code, and the drafting and revision
733 of both documents. Because the tools were used in combination and in alter-
734 nation, no section, proof, figure or result is attributable to any one of them,
735 and none is claimed to be. The author is responsible for all content, including

736 everything a tool produced. Machine checking establishes the stated Lean
737 results under their hypotheses; it does not validate biological interpretations
738 or replace scientific review of generated code and prose.

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